Data Structures

CSL202

Quiz Set-A

1. Given a string s containing only the characters (,), {, }, [,] determine if the input string is valid. A string is valid if: (a) Open brackets are closed by the same type of brackets. (b) Open brackets are closed in the correct order. (c) Every closing bracket has a corresponding opening bracket of the same type.

   Examples: (a) Input: s = ”()” Output: true

   (b) Input: s = ”()[]{}” Output: true

2. You are given an array of strings tokens that represents an arithmetic expression in Reverse Polish Notation (RPN) or Postfix expression. Evaluate the expression and return an integer that represents the value of the expression.

   Example 1:

   Input: tokens = [“2”, “1”, “+”, “3”, “*”]

   Output: 9

   Explanation: ((2 + 1) × 3) = 9

3. Design a stack that, in addition to push(x), pop(), and peek(), also supports retrieving the minimum element in constant time. Implement the following operations:

   (a) push(x) → Insert element x into the stack.

   (b) pop() → Remove the element from the top of the stack and return it. If the stack is empty, return -1.

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Quiz Set-B

1. Given a 0-indexed string word and a character ch, reverse the segment of word that starts at index 0 and ends at the index of the first occurrence of ch (inclusive). If the character ch does not exist in word, do nothing. Return the resulting string.

   Example 1: Input: word = "abcdefd", ch = "d" Output: "dcbaefd"

   Explanation: The first occurrence of "d" is at index 3. Reverse the part of word from 0 to 3 (inclusive), the resulting string is "dcbaefd".

2. You are given an array of strings tokens that represents an arithmetic expression in Reverse Polish Notation (RPN) or Postfix expression. Evaluate the expression and return an integer that represents the value of the expression.

   Example :

   Input: tokens = [“2”, “1”, “+”, “3”, “*”]

   Output: 9

   Explanation: ((2 + 1) × 3) = 9

3. The next greater element of some element x in an array is the first greater element that is to the right of x in the same array. You are given two distinct 0-indexed integer arrays nums1 and nums2, where nums1 is a subset of nums2. For each 0 <= i < nums1.length, find the index j such that nums1[i] == nums2[j] and determine the next

(c) peek() → Return the top element without removing it. If the stack is empty, return -1.

(d) getMin() → Return the minimum element in the stack. If the stack is empty, return -1.

Example:

Input: push(5), push(3), push(7), getMin(), pop(), getMin()

Output: 3, 3

Explanation:

Stack after pushes → [7, 3, 5] Minimum element = 3

After pop() (removes 7) → stack = [3, 5], minimum = 3

greater element of nums2[j] in nums2. If there is no next greater element, then the answer for this query is -1.

Return an array ans of length nums1.length such that ans[i] is the next greater element as described above.

Example 1:

Input: nums1 = [4,1,2], nums2 = [1,3,4,2]

Output: [-1,3,-1]

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Quiz Set-C

1. Given a string s containing only the characters (,), {, }, [,] determine if the input string is valid. A string is valid if: (a) Open brackets are closed by the same type of brackets. (b) Open brackets are closed in the correct order. (c) Every closing bracket has a corresponding opening bracket of the same type.

   Examples:

   (a) Input: s = ”()” Output: true

   (b) Input: s = ”()[]{}” Output: true

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Quiz Set-D

1. Given a 0-indexed string word and a character ch, reverse the segment of word that starts at index 0 and ends at the index of the first occurrence of ch (inclusive). If the character ch does not exist in word, do nothing. Return the resulting string.

   Example :

   Input: word = "abcdefd", ch = "d"

   Output: "dcbaefd"

   Explanation: The first occurrence of "d" is at index 3.

   Reverse the part of word from 0 to 3 (inclusive), the resulting string is "dcbaefd".

2. Implement a stack using a singly linked list

   peek() → Return the top element without removing it. If the stack is empty, return -1.

   Examples:

   Input: push(10), push(20), push(30), peek(), pop(), peek(), isEmpty()

   Output: 30, 30, 20, false

3. The next greater element of some element x in an array is the first greater element that is to the right of x in the same array. You are given two distinct 0-indexed integer arrays nums1 and nums2, where nums1 is a subset of nums2. For each 0 <= i < nums1.length, find the index j such that nums1[i] == nums2[j] and determine the next greater element of nums2[j] in nums2. If there is no next greater element, then the answer for this query is -1.

   Return an array ans of length nums1.length such that ans[i] is the next greater element as described above.

   Example 1: Input: nums1 = [4,1,2], nums2 = [1,3,4,2]

2. Implement a stack using a singly linked list

   peek() → Return the top element without removing it. If the stack is empty, return -1. Examples:

   Input: push(10), push(20), push(30), peek(), pop(), peek(), isEmpty()

   Output: 30, 30, 20, false

3. Design a stack that, in addition to push(x), pop(), and peek(), also supports retrieving the minimum element in constant time. Implement the following operations:

   (e) push(x) → Insert element x into the stack.

   (f) pop() → Remove the element from the top of the stack and return it. If the stack is empty, return -1.

   (g) peek() → Return the top element without removing it. If the stack is empty, return -1.

   (h) getMin() → Return the minimum element in the stack. If the stack is empty, return -1.

   Examples:

   Input: push(5), push(3), push(7), getMin(), pop(), getMin()

   Output: 3, 3

   Explanation:

   Stack after pushes → [7, 3, 5] Minimum element = 3

   After pop() (removes 7) → stack = [3, 5], minimum = 3